CoffyCup

A Premium Coffee Brand Website

A Project Report

Submitted by:

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in partial fulfillment for the award of the degree

of

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At



Department of Computer Science

INDUS UNIVERSITY

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ज्ञानेन प्रकाशते जगत्
INDUS UNIVERSITY

CERTIFICATE

This is to certify that the project titled "**CoffyCup**" is the bonafide work carried out by **Apoorv Batham**, a student of Integrated BCA - MCA SEM - V during the academic year 2026 - 27, in partial fulfillment of the requirements for the award of the degree of Bachelor of Computer Application.

Internal Guide Signature : **Head of Dept. Signature :**

Date : **Date :**

Acknowledgement

Since this was my first project, I was a little confused in the beginning about how to start and how to handle the challenges that might come along the way.

Throughout this project, my guide, **Dr. Shefali Parihar**, helped me a lot. Whenever I had any doubts or faced any difficulties, she guided me patiently and helped me understand things better. Her support and guidance helped me move forward with more confidence and complete my project.

I would also like to thank **Dr. Mukta Agrawal, Head of the Department**, for providing support and an encouraging learning environment. I am grateful to **Indus University** for giving me the opportunity to work on this project as part of my academic curriculum.

Finally, I am happy to present my project report titled **"CoffyCup – A Premium Coffee Brand Website."** Working on this project was a good learning experience and helped me gain practical knowledge of web development.

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Introduction

CoffyCup – A Premium Coffee Brand Website is a web-based project developed to showcase different coffee products, ready-to-drink beverages, and gifting hampers through an attractive and user-friendly interface. The main aim of this project is to provide an engaging online platform where users can explore coffee products and learn about the offerings of the CoffyCup brand.

The website is designed with a premium coffee-themed appearance to make the browsing experience simple, appealing, and convenient. It includes different sections for displaying products, user registration and login, and order-related functionality. The website also incorporates database integration to manage relevant information.

This project uses HTML, CSS, and JavaScript for the front-end interface, Python with Flask for the back-end development, and MySQL for database management. These technologies work together to provide an interactive website and support its core functionalities.

The development of CoffyCup provides practical experience in web development, database connectivity, and the integration of different technologies. It also helps in understanding how a product-based website can present its products and manage user-related information in an organized manner.

Existing system

Difficulty in Choosing Products: Traditionally, customers purchase coffee by visiting coffee shops or exploring online shopping platforms. However, general e-commerce websites offer a wide range of products and options, which can make it difficult for customers to focus on coffee products.

Lack of a Unified Platform: Coffee blends, ready-to-drink beverages, and gifting hampers may be spread across different sections or platforms. Customers may need to explore multiple places to find these products together, making browsing less convenient.

Need for a Coffee-Focused Experience: CoffyCup focuses specifically on coffee products. It brings coffee blends, ready-to-drink beverages, and gifting hampers together in one place while maintaining a coffee-inspired atmosphere. The website is designed not only to showcase coffee products but also to create a distinctive experience that reflects the feeling, warmth, and memories associated with coffee. It offers a carefully selected variety of products so that users can explore their options without feeling overwhelmed by too many choices.

Need for Better Product Presentation: A well-organized display of product names, prices, images, and descriptions can help customers understand the available options more easily.

Problem areas of existing system

Difficulty in Choosing Products: General e-commerce websites offer a wide variety of products, which can make it difficult for users to focus on coffee products and choose according to their preferences.

Lack of a Coffee-Focused Customer Experience: Traditional coffee shops and general e-commerce platforms mainly focus on selling products. General shopping platforms offer products from many categories and may not provide a **dedicated coffee-themed experience.** In contrast, CoffyCup aims to engage customers, create a special coffee-inspired atmosphere, and help them choose a suitable product without overwhelming them with too many options.

Limited Emotional Connection with Customers: Many shopping platforms focus mainly on product details and prices, giving less attention to the feelings and experiences associated with coffee. A coffee-focused website can create a more engaging atmosphere and help customers connect with the brand.

Difficulty in Exploring Different Coffee Categories: When coffee products, ready-to-drink beverages, and gifting hampers are not presented in a clear and organized way, users may find it difficult to explore different options. A well-structured website can make browsing easier and improve the overall user experience.

Need for new system

Dedicated Coffee Platform: A dedicated website is needed to bring different coffee blends, ready-to-drink beverages, and gifting hampers together in one place, making it easier for customers to explore coffee products.

Better Customer Experience: A coffee-focused website can provide an attractive and engaging browsing experience that reflects the feeling and atmosphere of coffee, rather than focusing only on product sales. CoffyCup aims to keep customers engaged by creating a special experience that connects coffee with emotions, special moments, and lifetime memories.

Easy Product Selection: A carefully selected range of coffee products can help customers explore available options and choose a suitable product without feeling overwhelmed by too many choices.

Order Status Tracking: Customers can view their orders and check their current status through the My Orders section, making it easier to keep track of their purchases.

Organized Product Information: Displaying product images, names, prices, and descriptions in an organized manner helps customers understand the available products and make informed choices.

PROPOSED SYSTEM

Coffee-Centric Experience: CoffyCup is a coffee brand website designed to provide customers with more than just a shopping experience. It aims to create a special coffee-themed atmosphere that engages customers and connects coffee with emotions, special moments, and lasting memories.

Unified Product Categories: The website brings coffee blends, ready-to-drink beverages, and gifting hampers together in one place.

Attractive and User-Friendly Interface: It offers a carefully selected variety of products through an attractive and user-friendly interface, helping customers explore and choose products easily.

Easy Online Ordering: Customers can select their preferred coffee products and place orders through the website.

Order Management: Customers can view their orders and check their order status through the My Orders section.

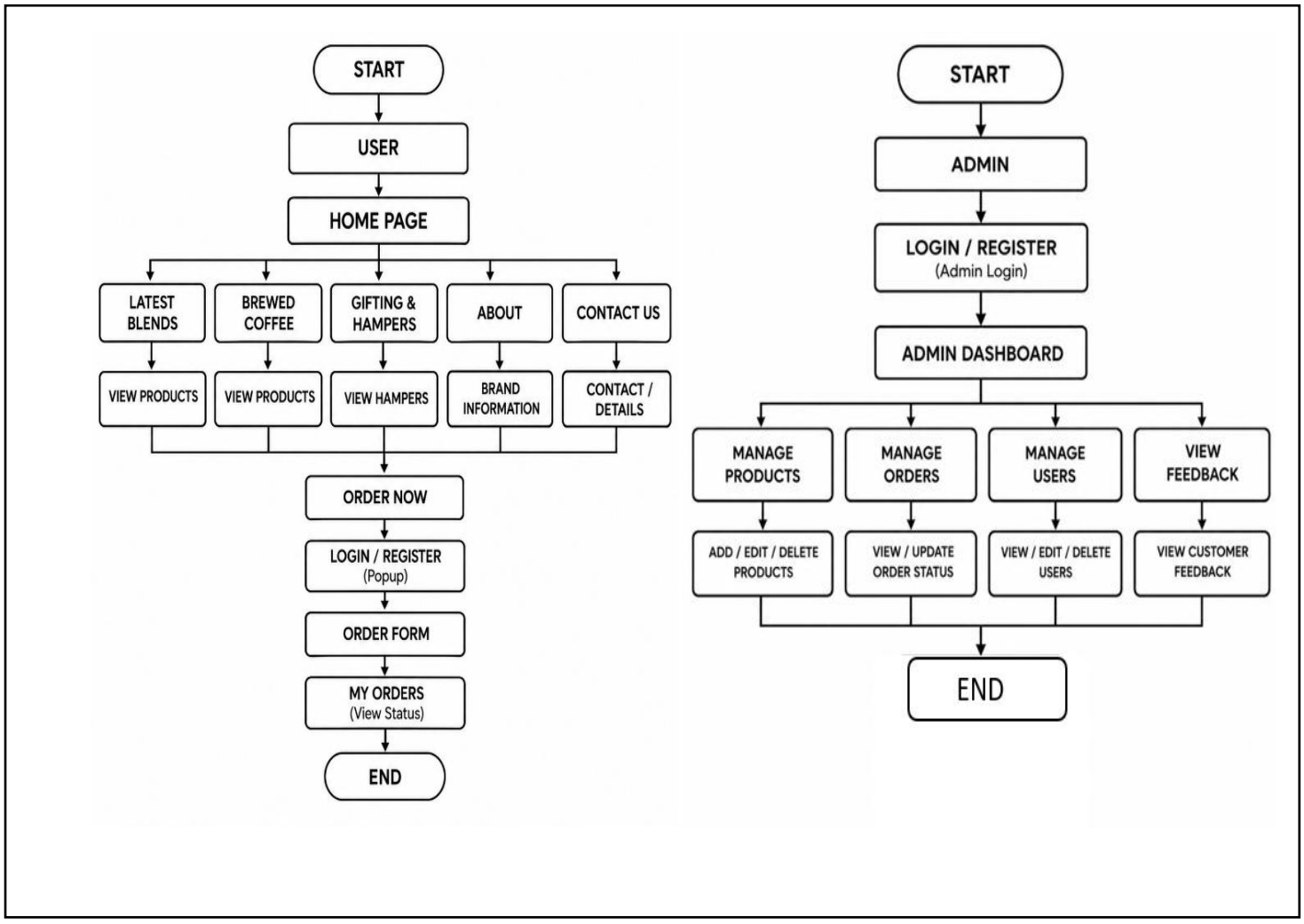
Admin Panel: The system provides an admin panel to manage products and orders and view customer feedback.

Customer Feedback Facility: Customers can share their feedback, allowing the website to collect their opinions and experiences.

Technology Used: The system is developed using HTML, CSS, JavaScript, Python (Flask), and MySQL.

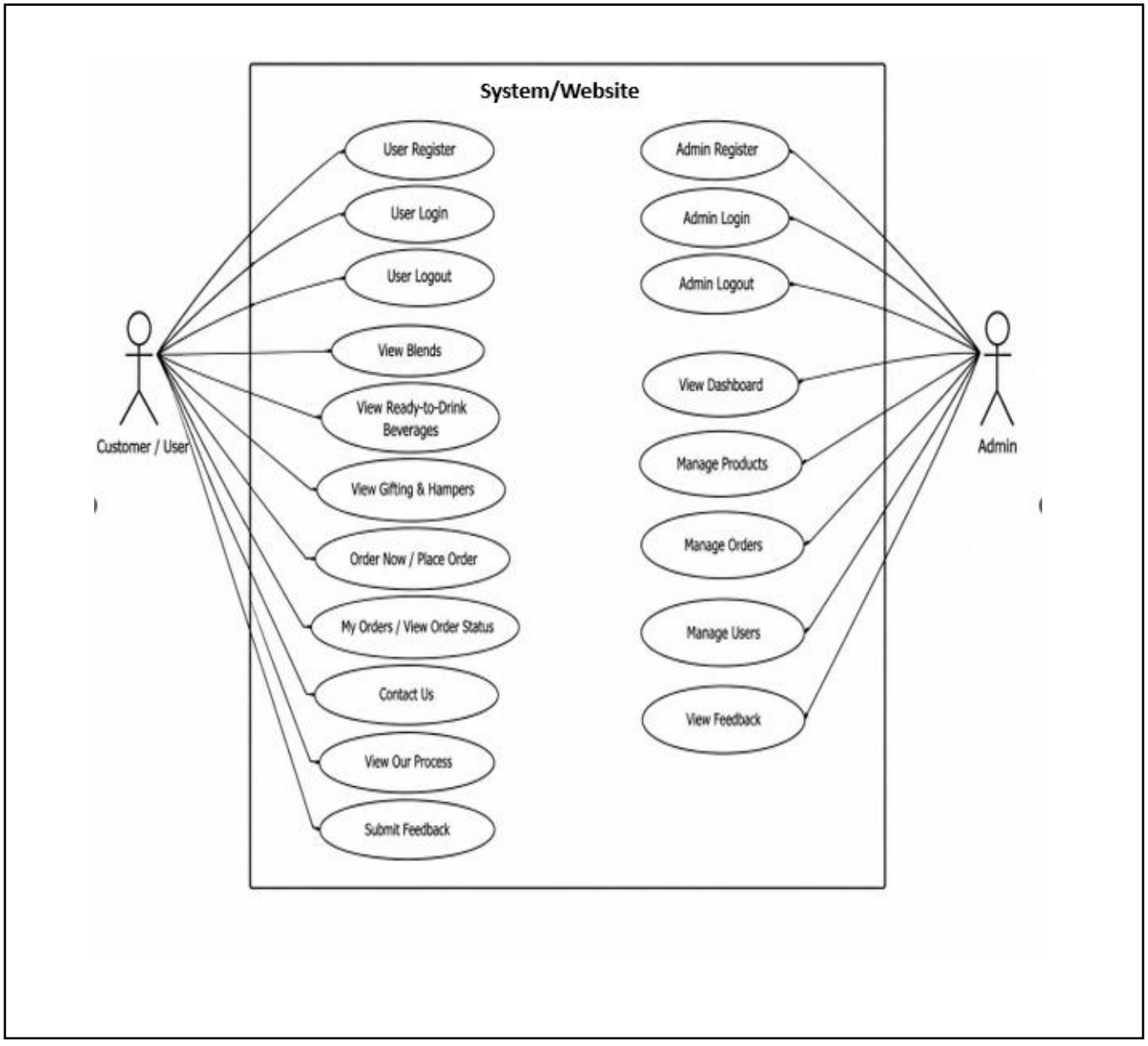
System Flow Diagram

User : **Admin:**



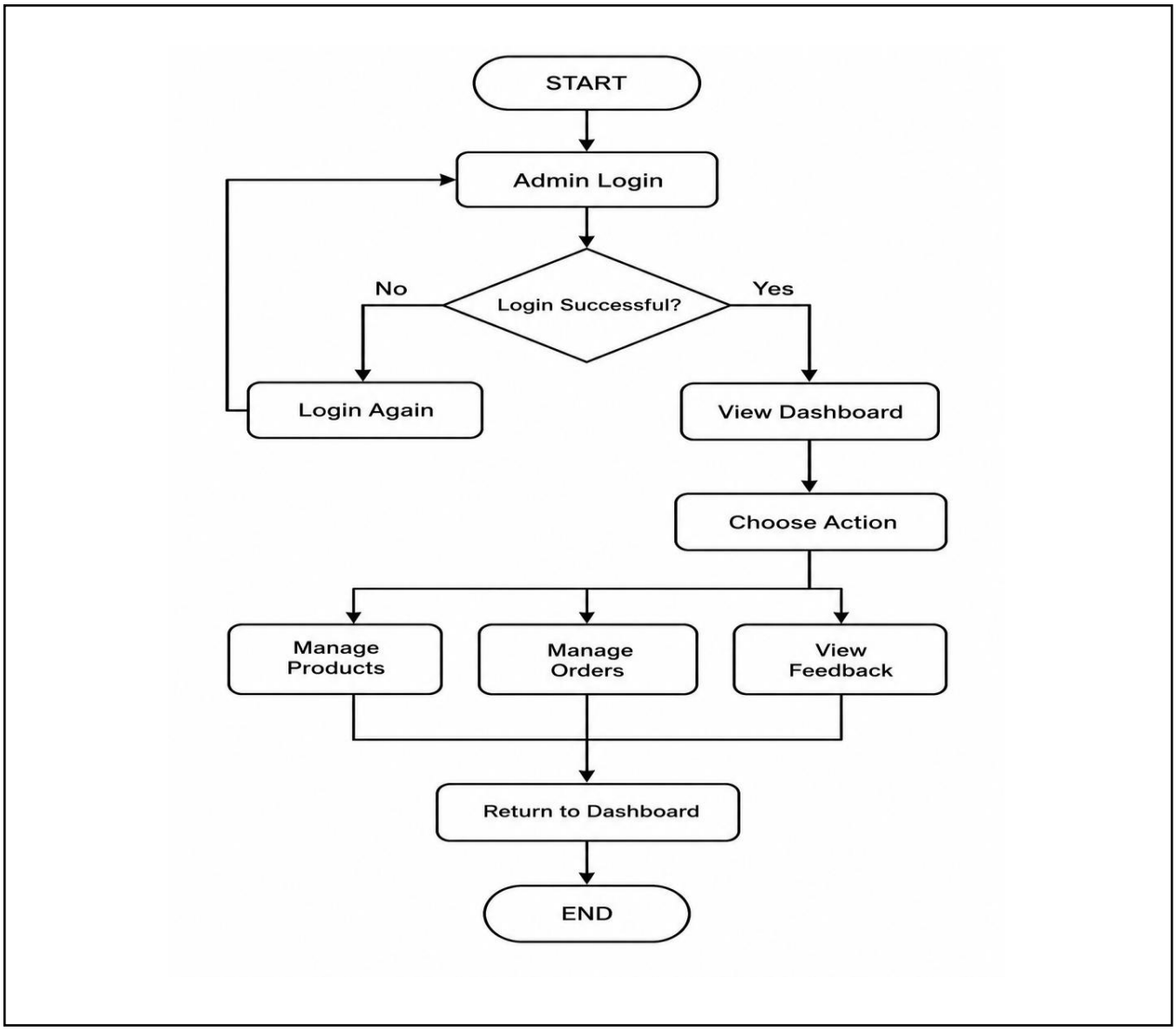
UML Diagram

1.Use Case Diagram :

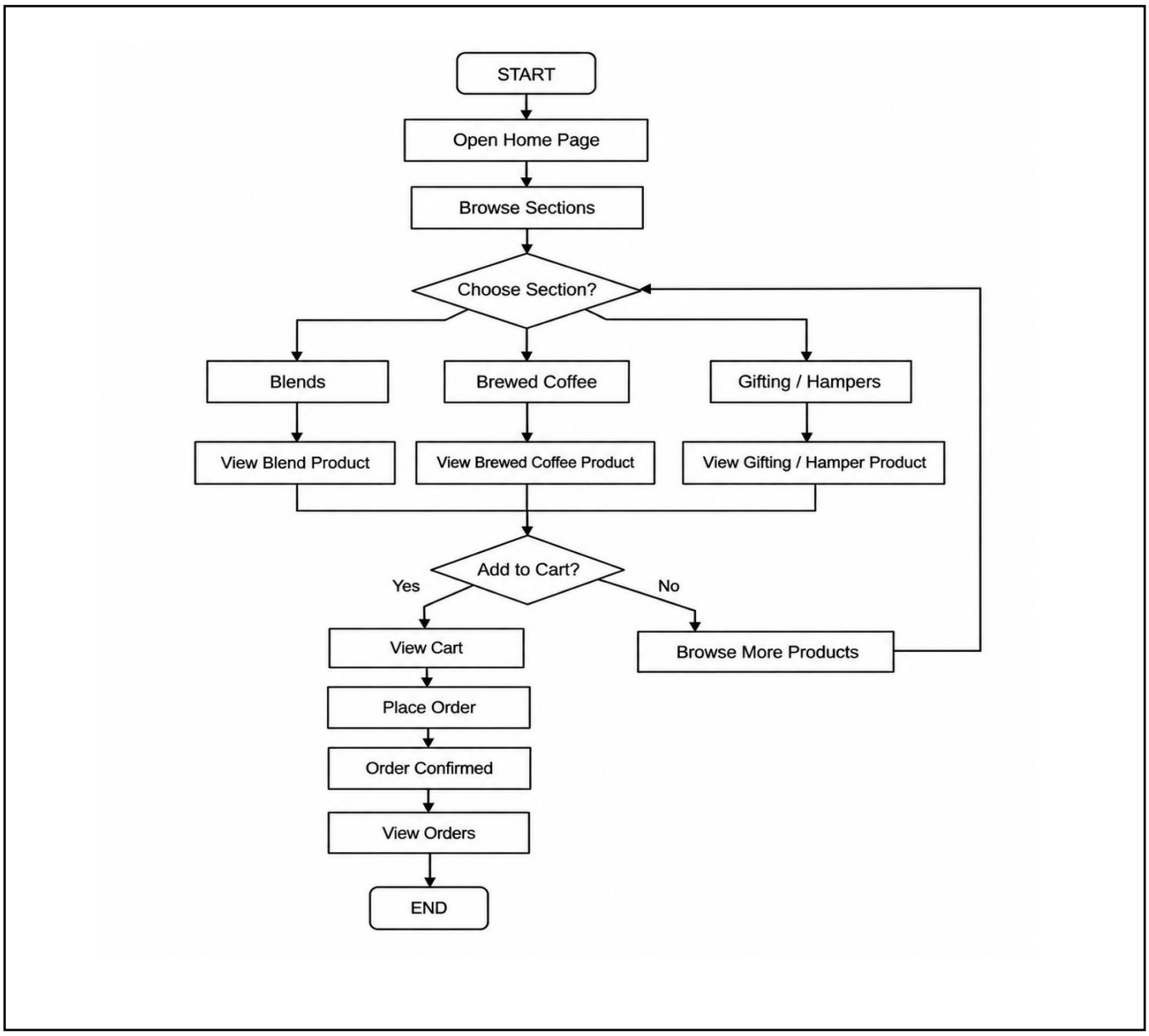


2.Activity Diagram :

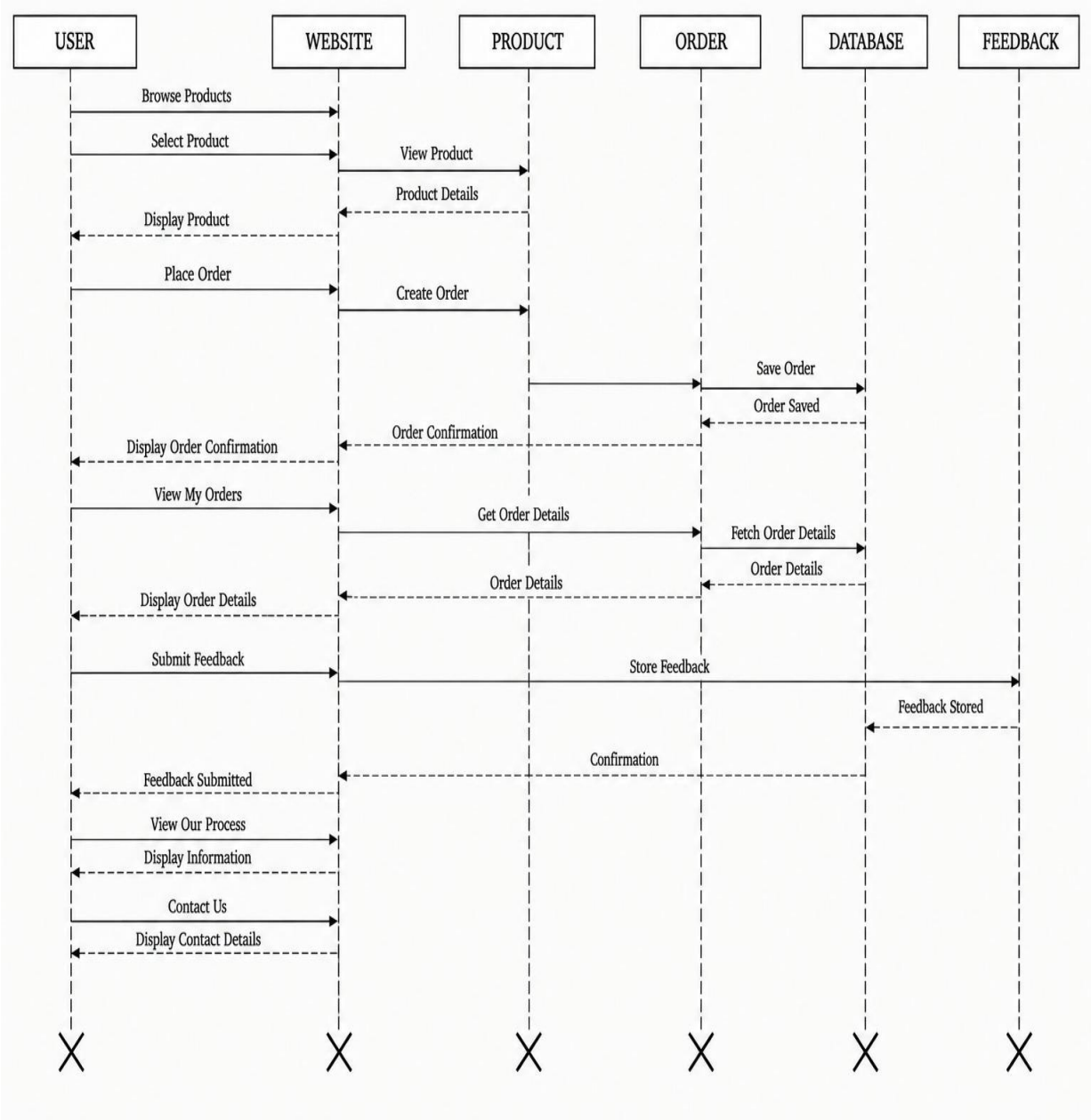
Admin



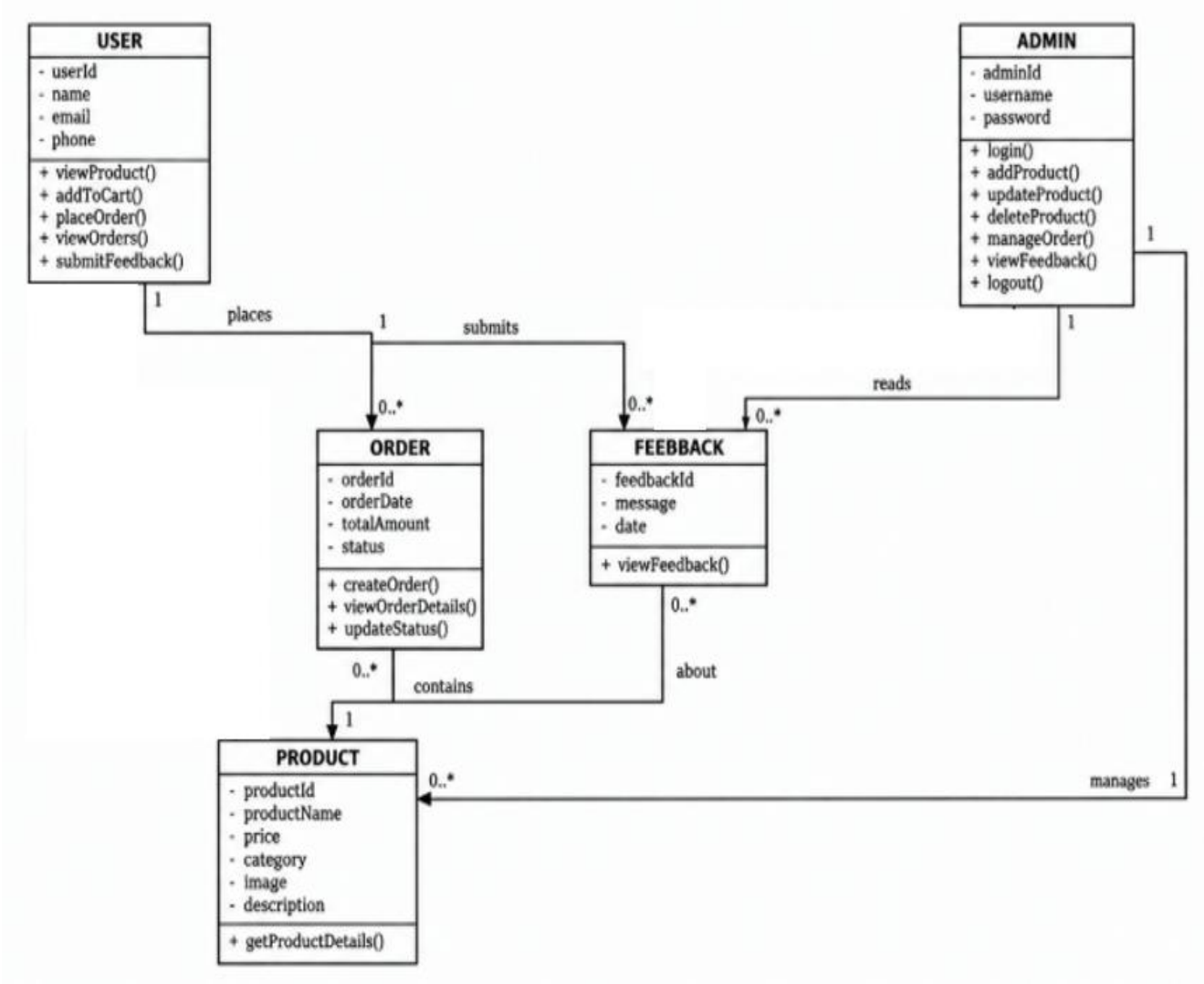
User


START
Open Home Page
Browse Sections
Choose Section?
Blends
Brewed Coffee
Gifting / Hampers
View Blend Product
View Brewed Coffee Product
View Gifting / Hamper Product
Add to Cart?
Yes
No
View Cart
Browse More Products
Place Order
Order Confirmed
View Orders
END

3. Sequence Diagram :



4. Class Diagram :



Data Dictionary

1. User

Field Name	Data Type	Constraints	Description
id	INT(11)	Primary Key, Auto Increment	Unique ID of each user.
name	VARCHAR(100)	NOT NULL	Name of the registered user.
email	VARCHAR(150)	NOT NULL, UNIQUE	Email address of the user.
phone	VARCHAR(20)	NULL Allowed	Contact number of the user.
password_hash	VARCHAR(255)	NOT NULL	Stores the hashed user password.
created_at	TIMESTAMP	NOT NULL, Default Current Timestamp	Date and time when the user registered.

2.Admin

Field Name	Data Type	Constraints	Description
id	INT(11)	Primary Key, Auto Increment	Unique ID of each admin.
username	VARCHAR(100)	NOT NULL, UNIQUE	Username used by the admin to log in.
password_hash	VARCHAR(255)	NOT NULL	Stores the hashed admin password.
created_at	TIMESTAMP	NOT NULL, Default Current Timestamp	Date and time when the admin account was created.

3. Product

Field Name	Data Type	Constraints	Description
id	INT(11)	Primary Key, Auto Increment	Unique ID of each product.
name	VARCHAR(150)	NOT NULL	Name of the coffee product or gifting hamper.
cat	VARCHAR(100)	NOT NULL	Category of the product.
price	DECIMAL(10,2)	NOT NULL	Price of the product.
img	VARCHAR(500)	Default Empty String	Path of the product image.
description	TEXT	NULL Allowed	Description of the product.
sold	INT(11)	NOT NULL, Default 0	Number of units sold, as recorded in the database.
created_at	TIMESTAMP	NOT NULL, Default Current Timestamp	Date and time when the product record was created.

3. Order

Field Name	Data Type	Constraints	Description
id	INT(11)	Primary Key, Auto Increment	Unique ID of each order.
user_id	INT(11)	Foreign Key, NULL Allowed	ID of the user associated with the order.
name	VARCHAR(100)	NOT NULL	Customer's name for the order.
phone	VARCHAR(20)	NOT NULL	Customer's contact number.
email	VARCHAR(150)	NULL Allowed	Customer's email address.
addr	TEXT	NOT NULL	Customer's delivery address.
city	VARCHAR(100)	NULL Allowed	Delivery city.
pincode	VARCHAR(15)	NULL Allowed	Delivery PIN code.
prod	VARCHAR(150)	NOT NULL	Name of the ordered product.
product_id	INT(11)	Foreign Key, NULL Allowed	ID of the related product.
qty	INT(11)	NOT NULL	Quantity ordered.
total	DECIMAL(10,2)	NOT NULL	Total amount of the order.
payment_method	VARCHAR(50)	Default Cash on Delivery	Payment method selected for the order.
status	VARCHAR(30)	NOT NULL, Default Pending	Current status of the order.
created_at	TIMESTAMP	NOT NULL, Default Current Timestamp	Date and time when the order was created.

4. Feedback

Field Name	Data Type	Constraints	Description
id	INT(11)	Primary Key, Auto Increment	Unique ID of each feedback entry.
user_id	INT(11)	Foreign Key, NULL Allowed	ID of the user who submitted feedback, if associated with an account.
name	VARCHAR(100)	NOT NULL	Name of the person submitting feedback.
email	VARCHAR(150)	NULL Allowed	Email address of the person submitting feedback.
msg	TEXT	NOT NULL	Feedback message submitted by the user.
created_at	TIMESTAMP	NOT NULL, Default Current Timestamp	Date and time when feedback was submitted.
rating	TINYINT(4)	NULL Allowed	Optional rating associated with the feedback.

User Interface

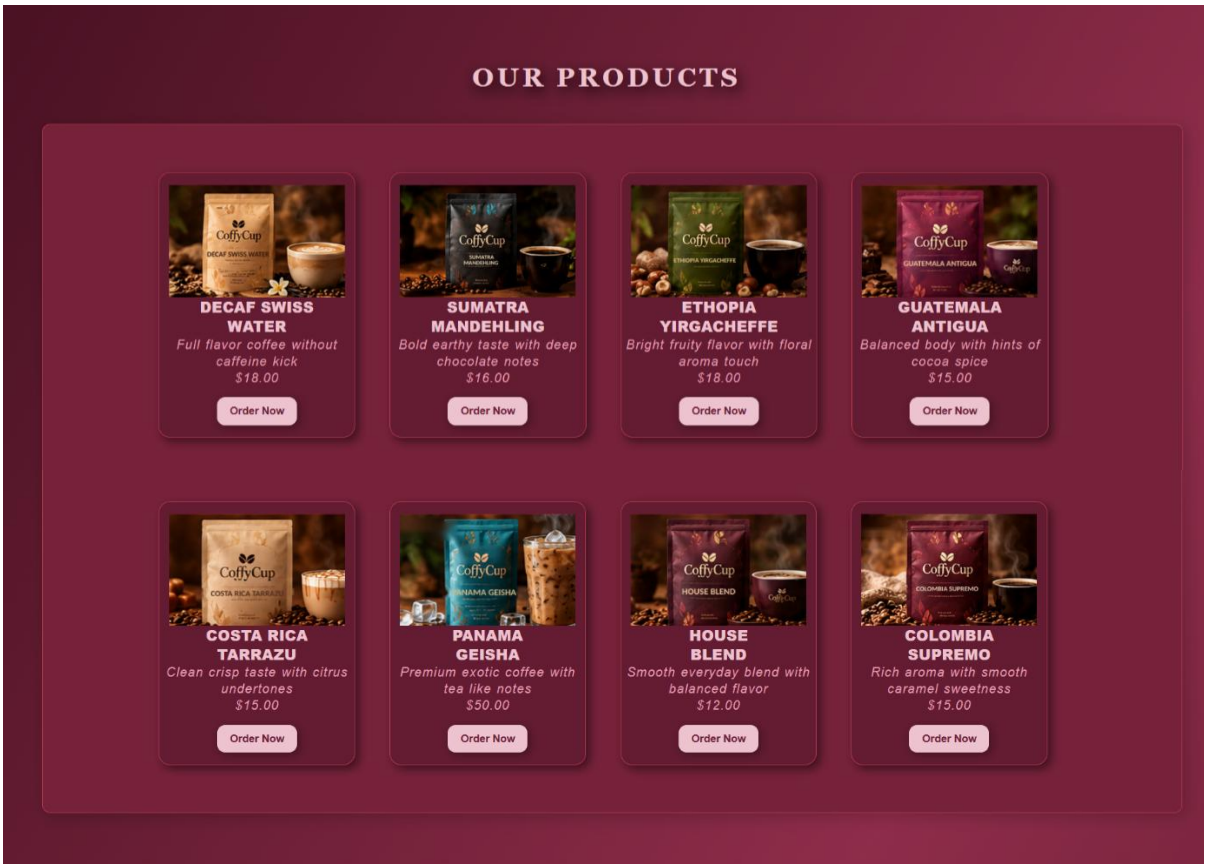
User-Side

- Home Page

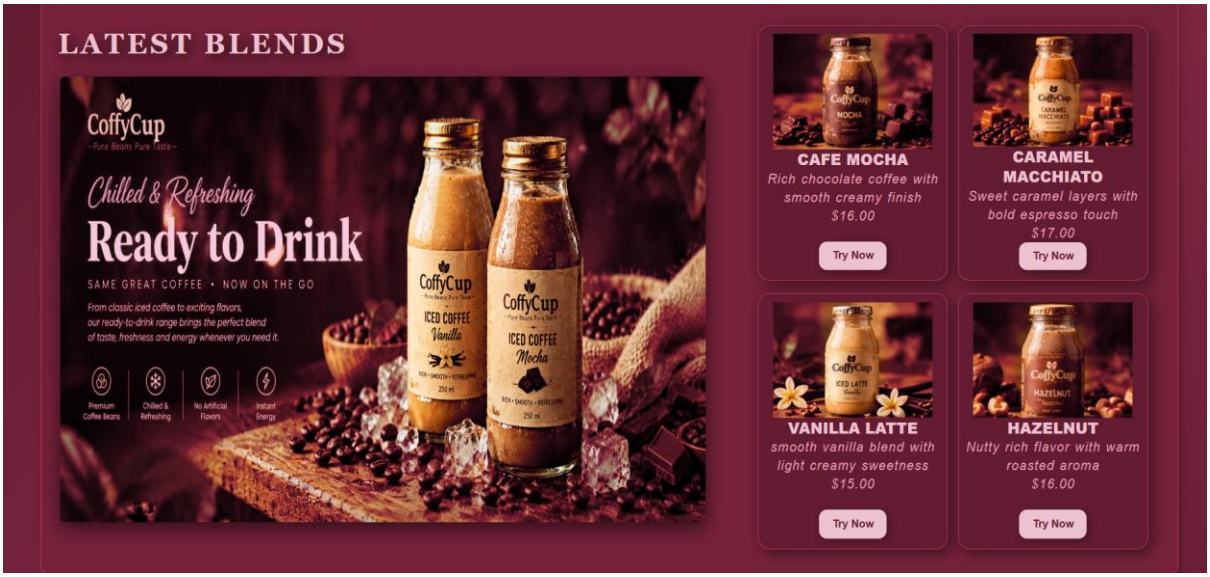


- Coffee Products

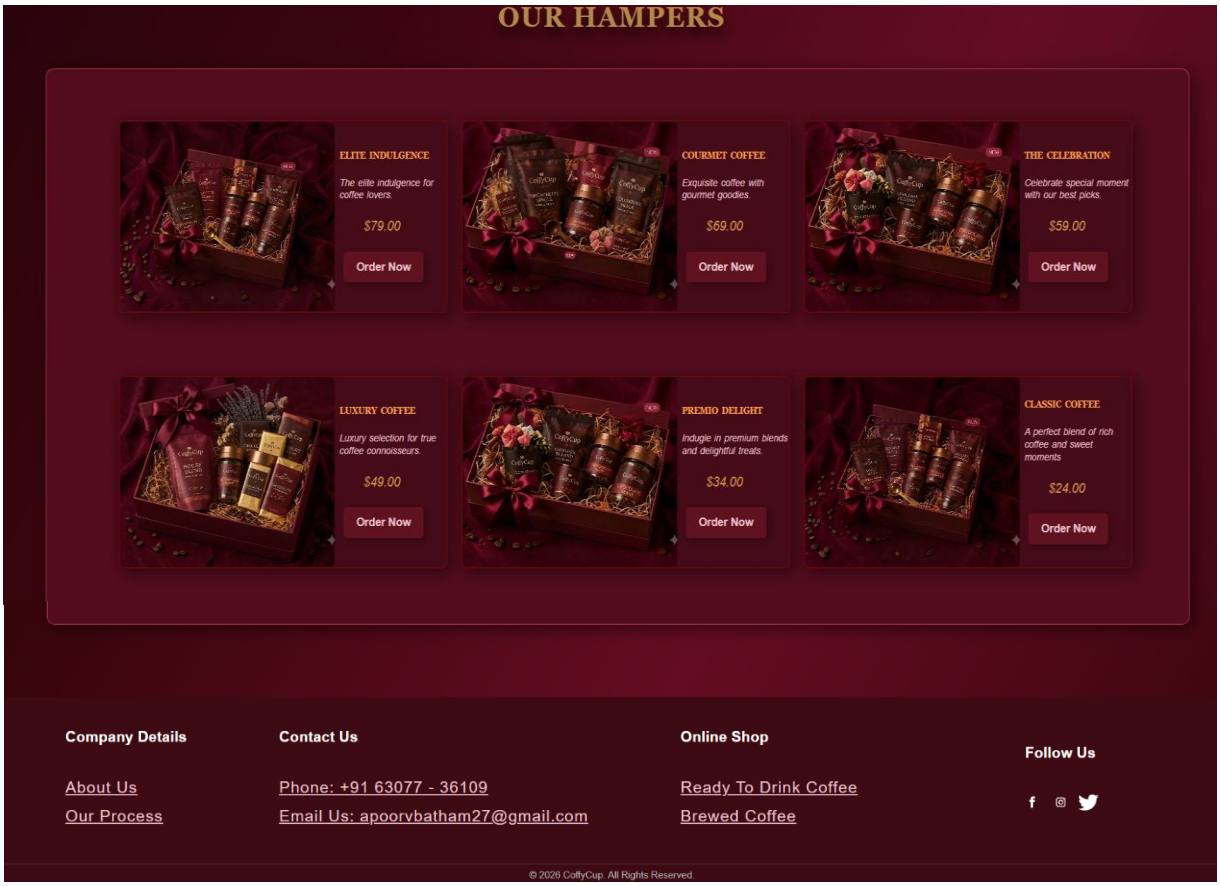
1.Coffee Blends:



2.Ready-to-Drink Beverages:

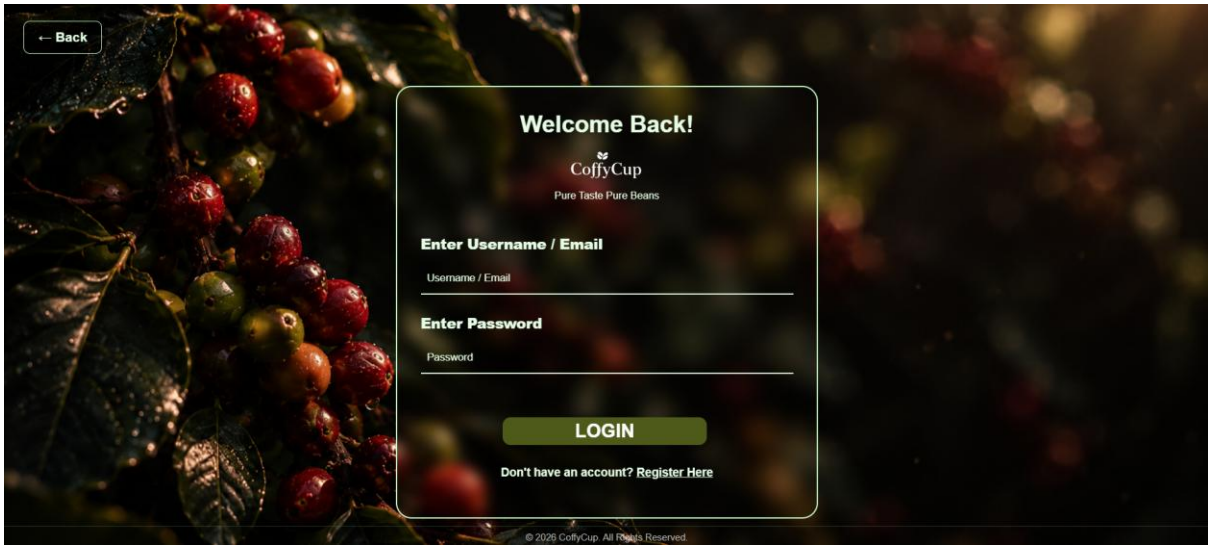


3.Gifting Hampers:

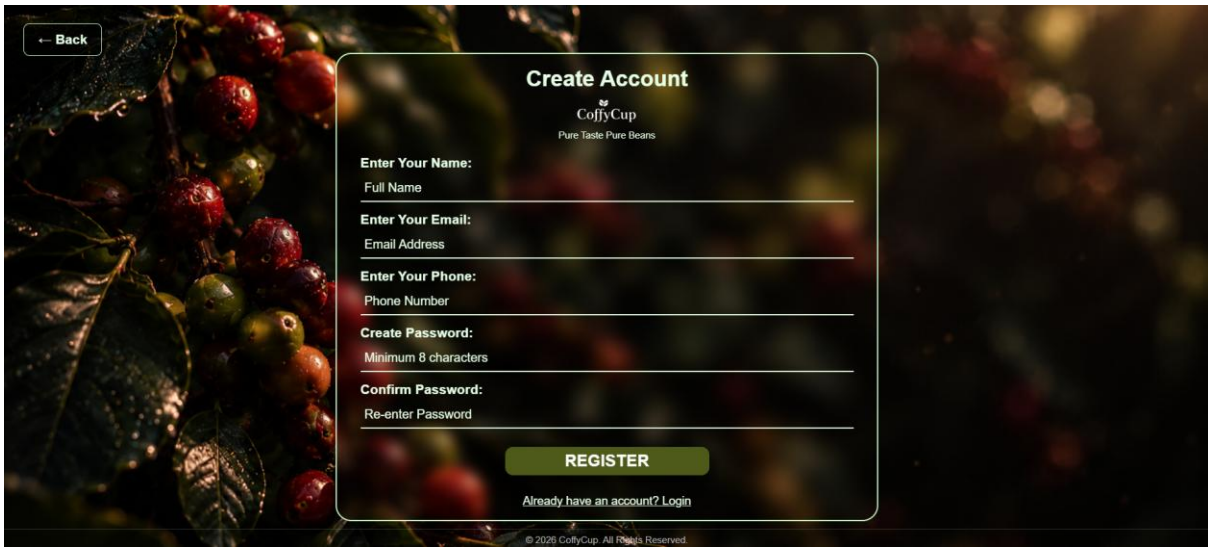


- Authentication

1.Login:

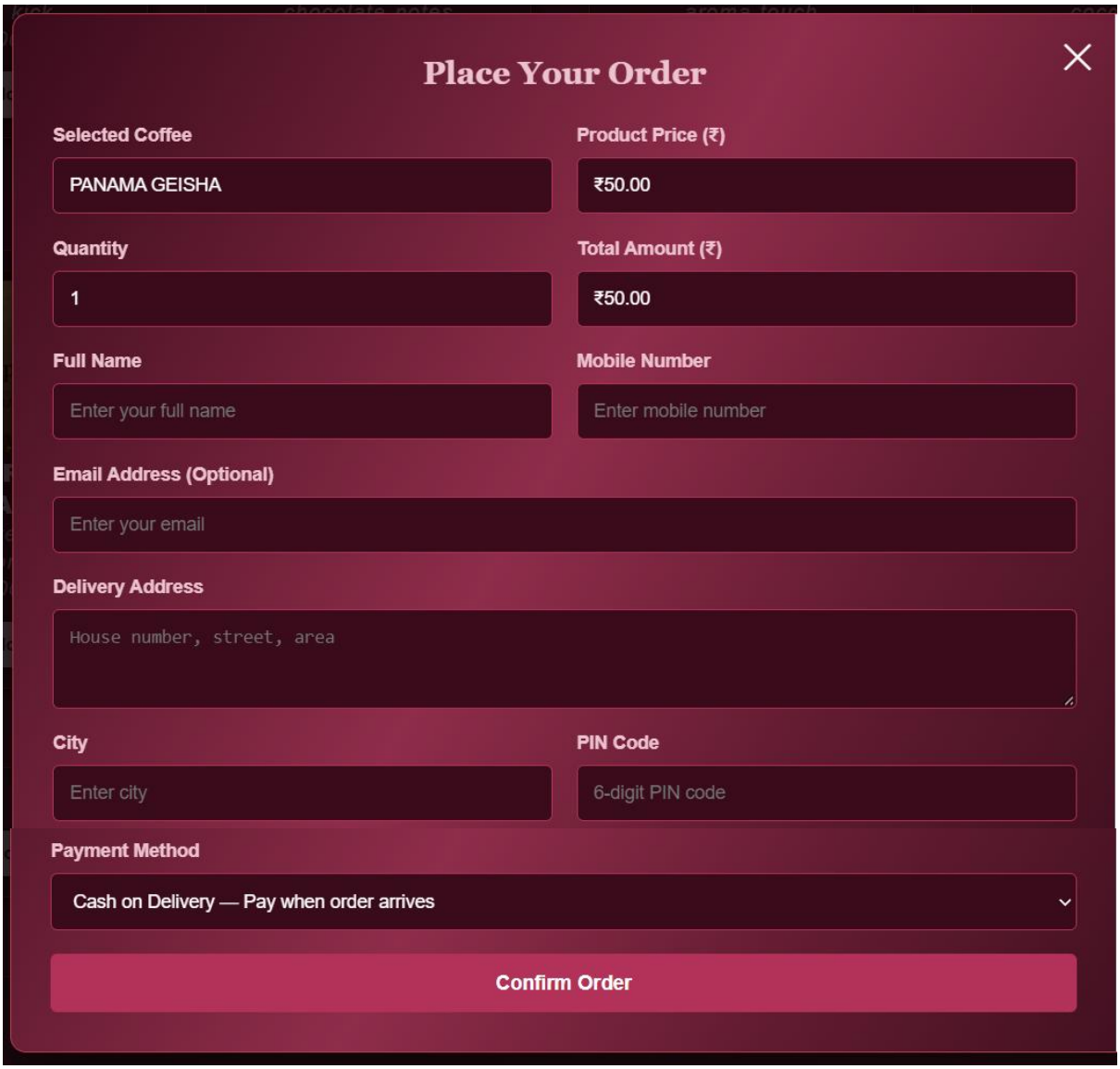


2.Register:

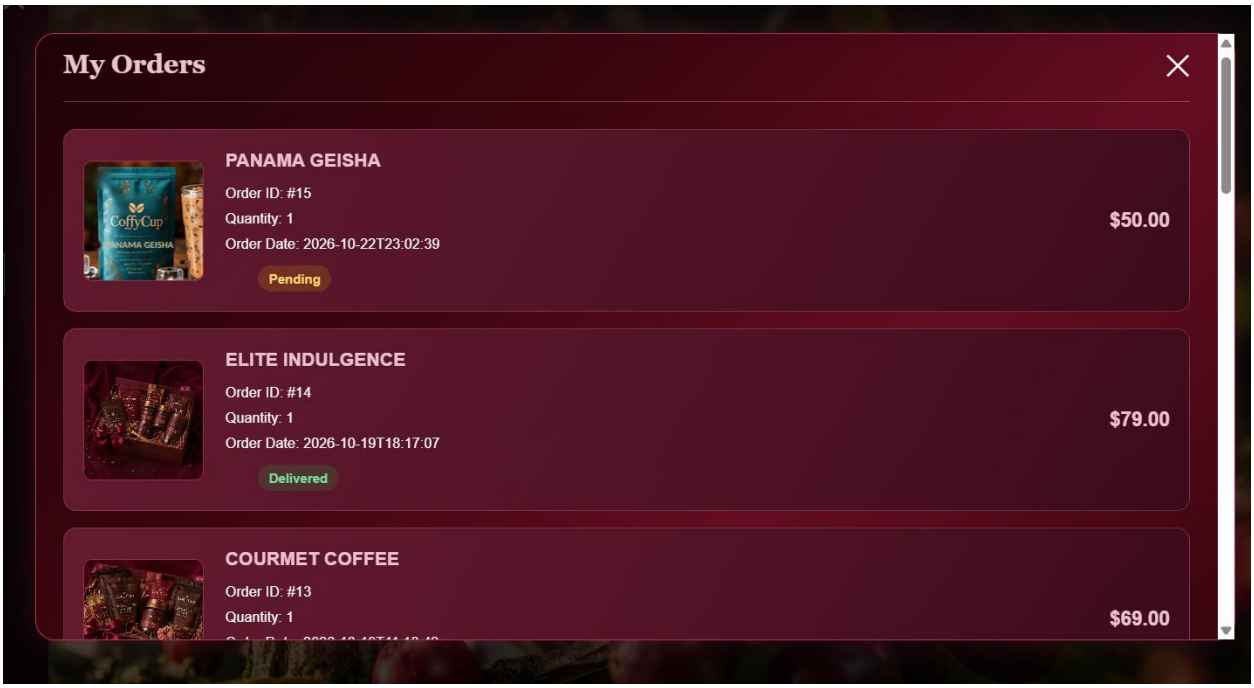


- Order Placement

1.Product-Selection

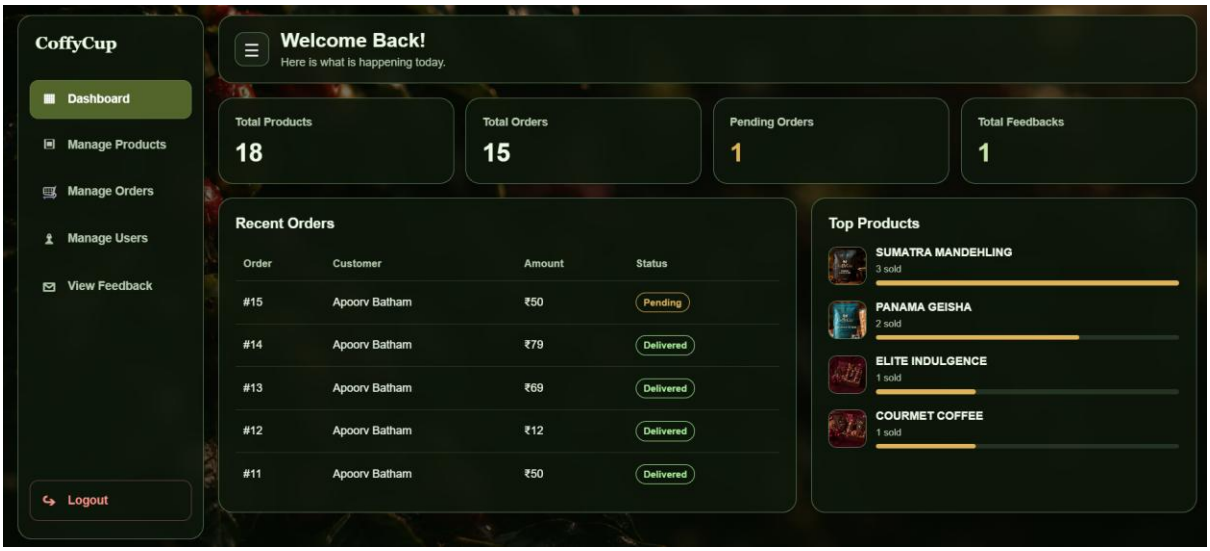


2.My-Cart

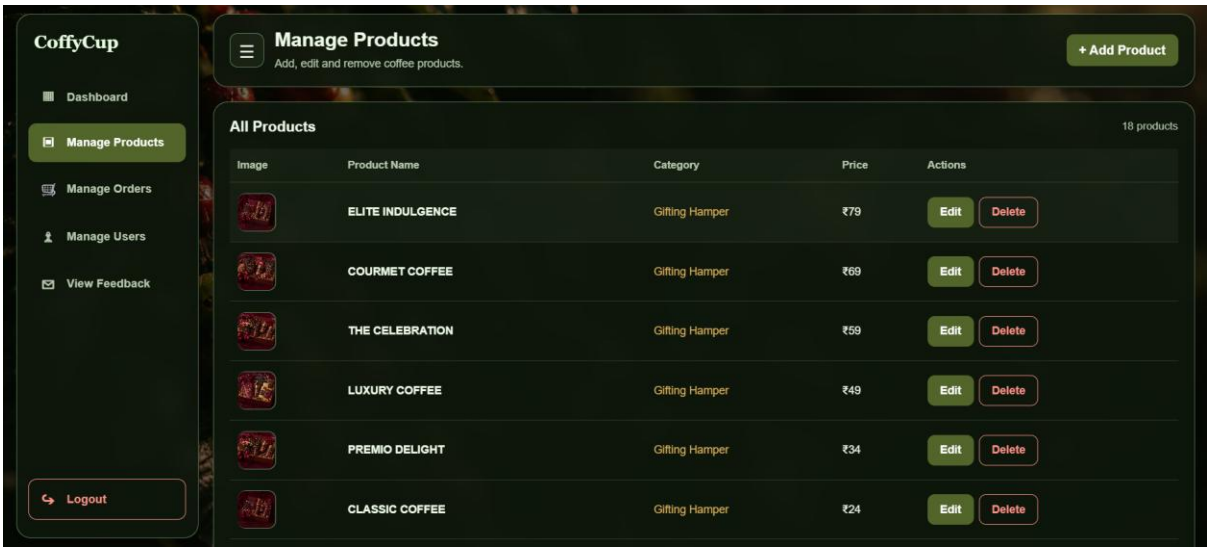

My Orders
PANAMA GEISHA
Order ID: #15
Quantity: 1
Order Date: 2026-10-22T23:02:39
Pending
$50.00
ELITE INDULGENCE
Order ID: #14
Quantity: 1
Order Date: 2026-10-19T18:17:07
Delivered
$79.00
COURMET COFFEE
Order ID: #13
Quantity: 1
$69.00

Admin-Side

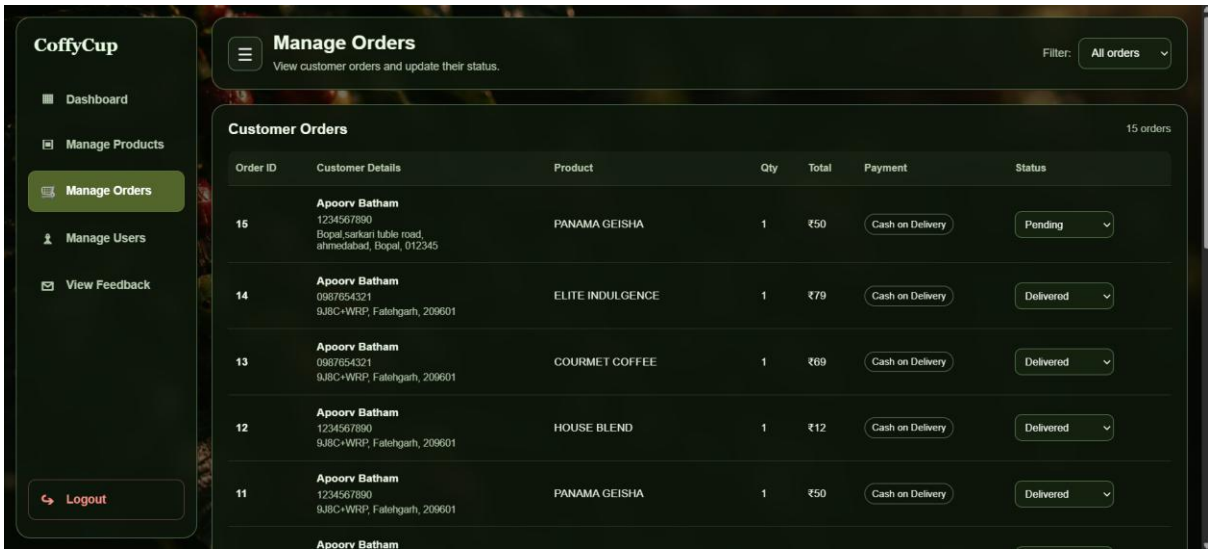
1.Admin Dashboard:



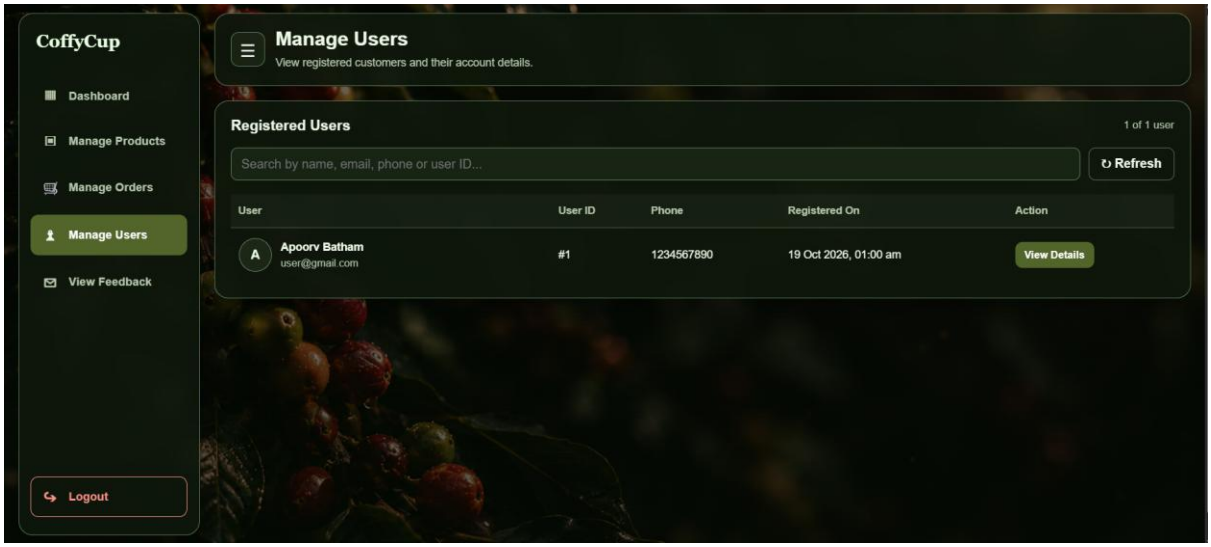
2.Products Management:



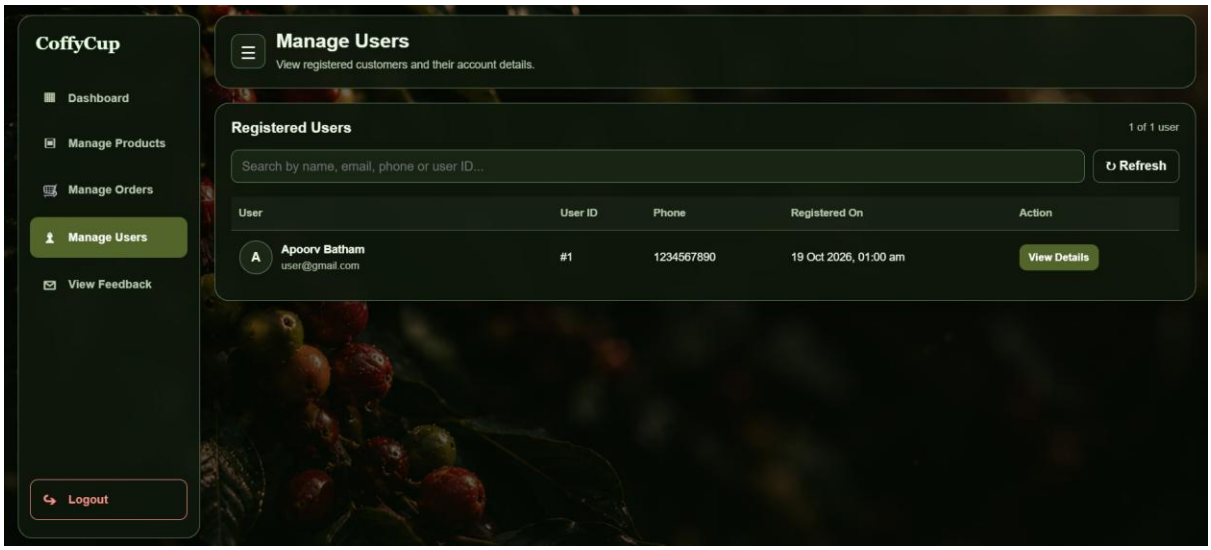
3.Order Management:



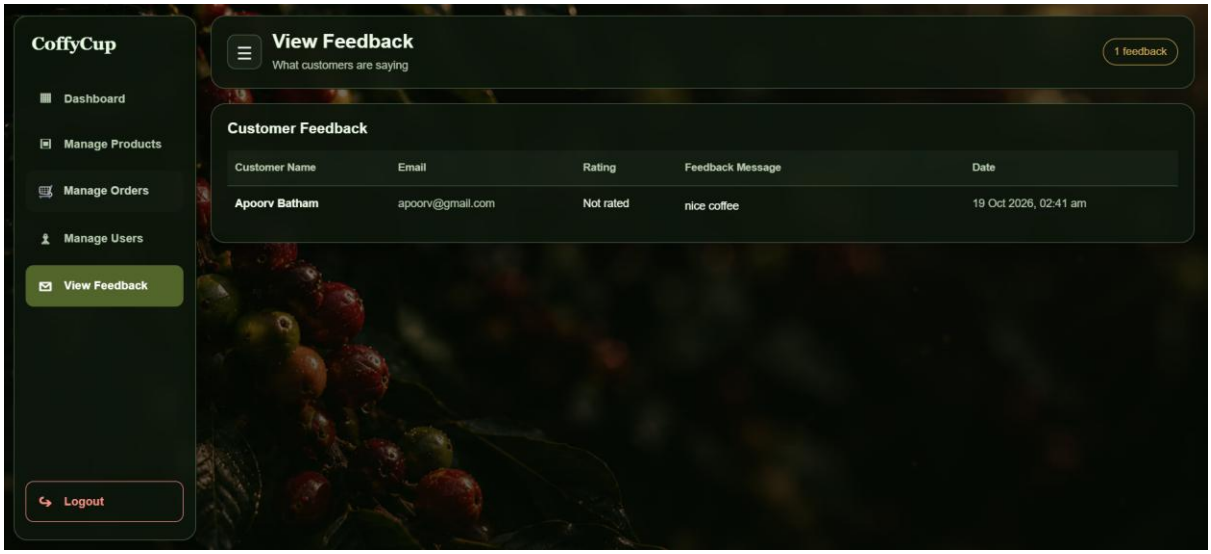
4.Order Management:



5.User Management:



5.View Feedback:



Test case & Test Plan

- **User-Side**

1. Test Case – Register Validation :

Test Case	**Test Case Description**	**Test Steps**	**Test Data**	**Expected Result**	**Actual Result**	**Pass/Fail**
1	**Registration Validation**	**Enter the required details and click Register.**	**Valid name, email and password**	**User account should be created successfully.**	**User account was created successfully.**	**Pass**
2	**Registration Validation**	**Leave required fields empty and click Register.**	**Empty required fields**	**Display a validation message for required fields.**	**Validation message was displayed.**	**Pass**

2. Test Case – Register Validation :

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
1	Login Data Verification	Enter valid email and password, then click Login.	Valid registered email and password	The user should log in successfully.	The user logged in successfully.	Pass
2	Login Data Verification	Enter incorrect credentials and click Login.	Invalid email or password	Display an error message for invalid credentials.	An error message was displayed.	Pass

3. Test Case – Blank Field Validation:

Test Case	**Test Case Description**	**Test Steps**	**Test Data**	**Expected Result**	**Actual Result**	**Pass/Fail**
1	Required Field Validation	Fill in all required fields and submit the form.	All required fields completed	The form should be submitted successfully.	The form was submitted successfully.	Pass
2	Empty Field Validation	Leave a required field empty and submit the form.	One or more required fields empty	The system should display a validation message.	A validation message was displayed.	Pass

4.Test Case – Product Selection

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
4	Coffee Product Selection	1. Open the Coffee Blends section. 2. Select a coffee product. 3. Click the Order Now button.	Selected coffee product	The order form should display the selected product.	To be filled after testing.	Pending

5. Test Case – Order Form Validation

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
1	Validate Order Details	Enter valid customer and delivery details.	Valid customer details and address	The form should accept valid details.	The form accepted valid details.	Pass
2	Validate Empty Order Fields	Leave required fields empty and attempt to confirm the order.	Missing required details	The system should prevent submission and display a validation message.	Submission was prevented and a validation message was displayed.	Pass

6. Test Case – Order Placement

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
1	Place an Order	Select a product, enter the required details and click Confirm Order.	Valid product and customer details	The order should be saved successfully.	The order was saved successfully.	Pass

7. Test Case – View My Orders

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
1	View Placed Orders	Log in and open My Orders.	User account with an existing order	The user's placed orders should be displayed.	The user's placed orders were displayed.	Pass

- **Admin Side**

1. Test Case – Admin Login Validation

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
1	Admin Login Validation	Open the admin login page, enter valid credentials, and click Login.	Valid admin username and password	The admin should log in successfully and access the dashboard.	The admin logged in successfully and accessed the dashboard.	Pass
2	Admin Login Validation	Enter invalid credentials and click Login.	Invalid username or password	An error message should be displayed.	An error message was displayed.	Pass

2. Test Case – Admin Dashboard

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
1	View Admin Dashboard	Log in as admin and open the dashboard.	Valid admin credentials	The dashboard should display the available statistics and information.	The dashboard displayed the available statistics and information.	Pass

3. Test Case – Manage Orders

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	**Pass/Fail**
1	View Customer Orders	Open the order management section.	Existing customer orders	The system should display the available customer orders.	The available customer orders were displayed.	Pass
2	View Order Details	Select an existing order to view its details.	Existing order details	The system should display the selected order details correctly.	The selected order details were displayed correctly.	Pass

4. Test Case – Manage Products

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
1	View Products	Open the product management section.	Available coffee products	The system should display the available products.	The available products were displayed.	Pass

5. Test Case – View Customer Feedback

Test Case	Test Case Description	Test Steps	Test Data	Expected Result	Actual Result	Pass/Fail
1	View Customer Feedback	Open the feedback section in the admin panel.	Existing customer feedback	The system should display the submitted feedback.	The submitted feedback was displayed.	Pass

Future Enhancement

- The CoffyCup website can be enhanced in the future by adding the following features:
- **Online Payment:** Integration of UPI and debit/credit card payment options.
- **Order Tracking:** Allow customers to check their order status.
- **Product Reviews and Ratings:** Enable users to rate and review coffee products.
- **Email Notifications:** Send order confirmations and status updates through email.
- **Mobile-Friendly Design:** Improve usability on mobile phones and tablets.
- **Wishlist:** Allow users to save their favourite coffee products for future reference.

References

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   https://www.w3schools.com/
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   https://docs.python.org/3/
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   https://www.drawio.com/
6. **Notepad++** – Code editing and website development.
   https://notepad-plus-plus.org/
7. **AI Tools** – Assistance with understanding concepts, resolving doubts, and improving project documentation and development.
   https://chatgpt.com/